[redacted]

Question - 1
Data Analysis Function

What is the output of the following code?

```
def analyze_data(data):
    if not data:
        return 'No data provided'
    elif isinstance(data, list) and all(isinstance(item, int) for item in data):
        return f'Max: {max(data)}, Min: {min(data)}'
    else:
        return 'Invalid data type'

print(analyze_data([10, 20, 30]))
```

- 'Max: 30, Min: 10'
- 'No data provided'
- 'Invalid data type'
- TypeError

Question - 2
Print Output

What is the output of the following?

```
print ("foo" * 3)
print (2 ** 3)
```

- foo foo foo
  2 2 2
- foofoofoo
  222
- foo foo foo
  8
- foofoofoo
  8

Question - 3

What is the expected output? 2

What is the expected output?

```
def foo(value):
    while True:
        value = (yield value)

bar = foo(1)
print(next(bar))
print(next(bar))
print(bar.send(2))
```

- 1
  1
- 1
  1
  2
- 1
  None
  2
- 1
  None
  None

Question - 4

Replacing Occurences

How can you replace all occurrences of the letter *a* with the letter *b* in a string variable named *example*?

- example.swap('a','b')
- example.replace('a','b')
- example.swap('b','a')
- example.replace('b','a')

Question - 5

What is the expected output? 1

What is the expected output?

```
b = [0,1,2,3,4,5,6,7,8,9]
print(b[::3])
```

- [2,5,8]

- [0,1,2,3]
- [7]
- [0, 3, 6, 9]

Question - 6
Rounding of a Negative Number

What is the output of the following?

```
ans = -4 // 1.5
print ans
```

- -2.0
- -3.0
- 2.0
- 3.0

Question - 7
Global Variable

How can we create a variable "foo" to be used globally across all modules.

- Create a module (config.py) and import it to all modules.
- Create a separate variable module (var.py), all variables in this file will automatically be imported to all modules.
- When declaring the variable, declare as a global variable with 'global'
- None of the above

Question - 8
What is required when creating a higher order function in Python?

What is required when creating a higher order function in Python?

- Nothing additional is required.
- Import the operfunc module.
- Import the functools module.
- It is not possible.

Question - 9

Copy a Compound Object

A _____ constructs a new compound object and then, recursively, inserts copies into it of the objects found in the original.

- Shallow copy
- Deep copy
- Simple copy
- Advanced copy

Question - 10

What will the following code do?

What will the following code do?

```
def foo(a,b,c): pass
```

- Defines an empty class foo
- Defines a list foo
- Defines a function, which does nothing
- None of the above

Question - 11

What is the difference between range() and xrange() in Python 2?

What is the difference between range() and xrange() in Python 2?

- range() returns an object and xrange() returns a list
- range() returns a list and xrange() returns an object
- xrange() does not exist and range() returns an list
- xrange() does not exist and range() returns an object

Question - 12

Operator Precedence in Python

What will strange(1230) return?

```
def strange(j):
  k = 0
  while j > 0:
    k = 10 * k + j % 10
    j = j // 10
  return k
```

- 0
- 123
- 321
- 1230
- 3210

Question - 13

How many return statements are allowed in a Python function?

How many return statements are allowed in a Python function?

- 0
- 1
- 0 or 1
- 1 or more
- Any number

Question - 14

Which of the following expressions causes a list indexing error?

Which of the following expressions causes a list indexing error?

- foo[-len(foo)]
- foo[-1]
- foo[0]
- foo[len(foo)]

Question - 15

What is the value of 'a' at the end of the loop?

What is the value of 'a' at the end of the loop?

```
a = 0
for b in range(0,10,2):
    a += b + 1
```

- 20
- 25
- 30

35

Question - 16
What is the value of "average" in the following code?

What is the value of "average" in the following code?

```
num1 = 100
num2 = 90
average = (num1 + num2) / 2.0
```

- 95
- 95.0
- 85
- 85.0

Question - 17
Which of the following statements will output:

Which of the following statements will output:

```
John
Charles
Chris
```

- print("John")
  print("Charles')
  print("Chris")
- print("John
  Charles
  Chris")
- print("John\nCharles\nChris")
- print("John\Charles\Chris")

Question - 18
Which of the following datatypes stores values in terms of key and value.

Which of the following datatypes stores values in terms of key and value.

- List
- Dictionary
- Class

- Tuple

Question - 19
What is the datatype of the following object?

What is the datatype of the following object?

```
obj = [1,15,"hello","30"]
```

- List
- Dictionary
- Array
- Tuple

Question - 20
What error message do you get when executing:

What error message do you get when executing:

```
hello = world
```

- SyntaxError
- NameError
- ValueError
- TypeError

Question - 21
String Slicing in Python

What is the output of the following?

```
>>>word = "positive"
>>>word[:2]
```

- positive
- po
- ve
- None of the above

Question - 22
Which of the following is a valid statement. (Multiple answers are possible)

Which of the following is a valid statement. (Multiple answers are possible)

- round(1234.1234)
- round(413.123.2)
- round(222.123,2)
- round()